C9 — The fair price, forecasts of the mid, and denoising on every asset

Not a ladder phase. It answers two questions asked after C8, and two follow-ups those answers raised. Lab notebook: `experiments/c9_fair_price/README.md`.

> **At a glance**
>
> - **Knowing the mid 1–3 s ahead is worth a lot, on most assets.** Quoting around the future mid instead of the current one raises held-out return on seven of eight configurations at 1 s and all eight at 3 s, and more the further ahead: ETH GLFT goes from −3.25% to +4.30% (1 s) and +19.67% (3 s), BNB GLFT from +1.72% to +14.72% (1 s). On XRP, whose quotes sit far outside a second's move, it changes little.
> - **The mid cannot be forecast well enough to use that.** The best causal forecast (boosted trees on 17 book, flow and momentum features) explains 1.8–10.2% of the 1 s move held out, depending on the asset. A forecast has to explain about 40% before centring the quotes on it beats the C8 choice on both training and held-out sessions.
> - **Every real fair price chosen on training sessions is a wash or worse held out, except on AVAXUSDC** (+3.33% against +2.68%, surviving the stress tests). The micro-price, a fitted micro-price, a linear and a boosted-tree forecast were each tried on eight configurations.
> - **Acting only on large moves would be enough — with a perfect detector.** Pulling a quote before the top 1% of 3 s moves takes ETH OBI from +0.85% to +6.53% held out. The forecasts' largest calls are about 0.6 bp against 3–5.5 bp real large moves, and the rules built on them do not help.
> - **DAgger with a privileged teacher** — a student that learns, from a teacher who sees the future mid, which quotes to pull — improves the losing ETH GLFT and does not help ETH OBI. With either teacher the GLFT student cuts the held-out loss from −3.25% to about −1.1% (19 fills a day against 27), and it still loses; on ETH OBI the students give +0.91% and +0.52% against +0.85%. The large-move teacher's student ranks those moves with an AUC of 0.71 but most of its pulls are false alarms.
> - **Denoising, now on all nine assets, does not help.** No denoiser raises held-out return on a configuration that makes money without failing a stress test; on SUIUSDC OBI every denoiser that helps training costs 1–3 points held out.

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1. The task

The request, after C8:

> how we can better calc mid-price (or fair price) from which we put spread — can we predict it over 1 s interval or so with high precision? (you can verify that if we will put spread from price with future lag around 0.5–1.5 s it will increase pnl) and lets test denoising across all assets and most profitable strategies at the moment

Then, once the first answers were in, two follow-ups: **check whether acting only on large moves is enough** (a "large-move detector" instead of a forecast of every move), and **try RL on it — DAgger with a teacher that sees the future**.

Four questions, then:

1. How well can the mid 0.5–3 s ahead be forecast from what the quoter sees?
2. What is perfect foresight worth — quoting around the future mid itself?
3. How good must a forecast be before quoting around it pays, and do the real ones get there?
4. Does denoising the signals help any profitable configuration, on any asset?

2. How it was run

C8's month split and cache: **24 training sessions** (2025-08-26 to 09-18), **7 held out** (09-19 to 09-25), every configuration chosen on training sessions, reported held out and stress-tested with C8's stress set. A choice must trade at least 20 times on a median training session. Results are cumulative return in % of the book.

The quoting kernel takes a new input: **one value per 100 ms quoting step, in bp of the mid**, which moves the centre of the quotes (added to the strategy's own signal, or replacing it), or widens or pulls only the side a coming move would run over. It is off by default, and every cached C8 session replays to the identical return and fill count with it compiled in.

The series come from a **feature file per session** (`features.py`): the same backtest the strategies run on, replayed with no orders, recording the book and flow after each 100 ms step exactly as the strategy sees them. The mid in that file matches the strategy's own mid on 100% of steps, so step `t` of a forecast is step `t` of the quotes — no timestamp matching, and no look-ahead beyond what the oracle means to have.

Configurations: **OBI with the markout rule on all nine assets and BNBUSDC GLFT** (everything that makes money held out, plus the rest of OBI) for denoising; for the fair price, the five OBI configurations that make money (ETH, XRPUSDC, XRPUSDT, SUIUSDC, AVAXUSDC), BNB GLFT, and ETH GLFT and hfgrid (which lose, to see whether a better centre rescues them). Each starts from its C8 choice.

3. Can the mid be forecast 0.5–3 s ahead?

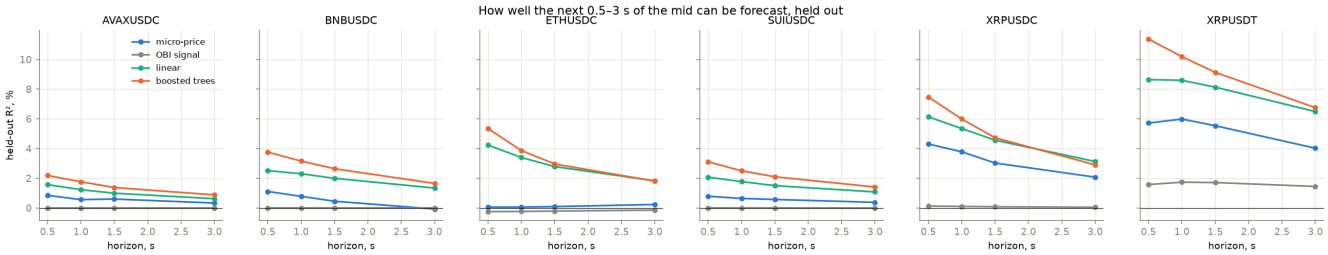


Figure — held-out R² of each forecast of the mid's move, by horizon.

Four forecasts, fitted on the training sessions and scored on every step of the held-out ones: the **micro-price**'s offset from the mid; **OBI's own signal** (the 10 bp band imbalance standardised over ten minutes, as the strategy uses it); a **linear** model and **boosted trees** (LightGBM, small and regularised) on 17 causal features — touch and band imbalances, OFI and trade-flow imbalance over 0.1–10 s, the mid's own recent returns, the spread.

asset	horizon	move std, bp	moves	micro-price	OBI signal	linear	boosted trees	trees, training	trees hit rate
AVAXUSDC	0.5 s	1.17	51%	0.8	−0.0	1.6	**2.2**	4.3	59%
AVAXUSDC	1 s	1.64	66%	0.6	−0.0	1.2	**1.8**	3.7	58%
AVAXUSDC	1.5 s	2.02	74%	0.6	−0.0	1.0	**1.4**	3.0	57%
AVAXUSDC	3 s	2.87	85%	0.3	−0.0	0.6	**0.9**	2.2	55%
BNBUSDC	0.5 s	0.67	35%	1.1	0.0	2.5	**3.8**	6.1	69%
BNBUSDC	1 s	0.96	50%	0.8	0.0	2.3	**3.2**	5.9	65%
BNBUSDC	1.5 s	1.19	60%	0.5	0.0	2.0	**2.6**	5.5	62%
BNBUSDC	3 s	1.71	77%	−0.1	0.0	1.3	**1.7**	4.6	59%
ETHUSDC	0.5 s	0.60	34%	0.1	−0.2	4.2	**5.4**	7.6	71%
ETHUSDC	1 s	0.88	49%	0.1	−0.2	3.4	**3.9**	5.6	67%
ETHUSDC	1.5 s	1.10	59%	0.1	−0.2	2.8	**3.0**	4.3	64%
ETHUSDC	3 s	1.61	74%	0.2	−0.1	1.8	**1.8**	2.7	60%

asset	horizon	move std, bp	moves	micro-price	OBI signal	linear	boosted trees	trees, training	trees hit rate
SUIUSDC	0.5 s	0.84	43%	0.8	0.0	2.1	**3.1**	5.3	60%
SUIUSDC	1 s	1.20	58%	0.6	0.0	1.8	**2.5**	4.3	58%
SUIUSDC	1.5 s	1.47	66%	0.6	0.0	1.5	**2.1**	3.5	57%
SUIUSDC	3 s	2.11	79%	0.4	−0.0	1.1	**1.4**	2.1	56%
XRPUSDC	0.5 s	0.59	29%	4.3	0.1	6.1	**7.5**	9.1	75%
XRPUSDC	1 s	0.87	43%	3.8	0.1	5.3	**6.0**	7.9	70%
XRPUSDC	1.5 s	1.09	53%	3.0	0.1	4.6	**4.7**	6.6	67%
XRPUSDC	3 s	1.58	68%	2.1	0.0	3.1	**2.9**	4.6	62%
XRPUSDT	0.5 s	0.55	20%	5.7	1.6	8.6	**11.4**	11.7	85%
XRPUSDT	1 s	0.82	31%	6.0	1.7	8.6	**10.2**	10.1	80%
XRPUSDT	1.5 s	1.04	39%	5.5	1.7	8.1	**9.1**	9.2	76%
XRPUSDT	3 s	1.55	55%	4.0	1.4	6.5	**6.8**	6.9	70%

Held-out R^2, %. "moves": share of steps on which the mid moves at all over the horizon. "hit rate": direction right, on steps where both the mid and the forecast move.

- **The best forecast explains 2–10% of the 1 s move**: 10.2% on XRPUSDT, 6.0% on XRPUSDC, 3.9% on ETHUSDC, 1.8% on AVAXUSDC. It gets the direction right 58–80% of the time — useful, but far from "high precision".
- **Shorter is easier**: every asset's R^2 falls from 0.5 s to 3 s.
- **The micro-price is the whole story on XRP** (3.8% of 6.0% at 1 s), where the spread is one wide tick, and **worth nothing on ETH** (0.1%), where the tick is 0.02 bp and the micro-price barely differs from the mid.
- **OBI's own signal forecasts nothing** ($R^2 \approx 0$, and below zero on ETH). Whatever OBI earns, it does not earn it by predicting the mid.

4. What perfect foresight is worth

The oracle quotes around the mid `k` steps ahead — the actual future mid, from the feature file — added to the strategy's own signal. It is not a strategy; it is the ceiling on what knowing the next seconds is worth.

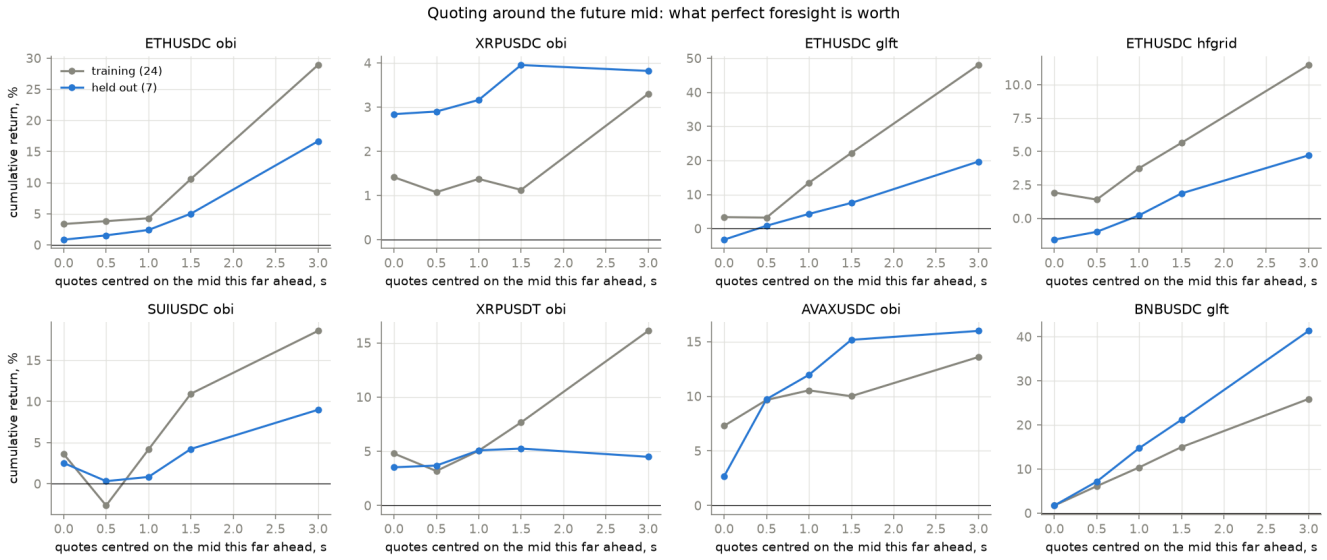


Figure — cumulative return when the quotes are centred on the mid this far ahead; 0 is the C8 choice.

configuration	sessions	as chosen	mid +0.5s	mid +1s	mid +1.5s	mid +3s	mid +1s, own signal off
ETHUSDC OBI	held out	+0.85	+1.54	+2.41	+5.02	+16.65	+2.16
ETHUSDC OBI	training	+3.37	+3.82	+4.29	+10.60	+28.89	+6.25
	fills/day held out	369	418	611	689	1321	10
XRPUSDC OBI	held out	+2.84	+2.90	+3.16	+3.95	+3.82	+2.55
XRPUSDC OBI	training	+1.41	+1.07	+1.37	+1.13	+3.30	+0.88
	fills/day held out	106	93	94	90	160	0
ETHUSDC GLFT	held out	−3.25	+0.82	+4.30	+7.53	+19.67	–
ETHUSDC GLFT	training	+3.37	+3.21	+13.44	+22.19	+48.01	–
	fills/day held out	27	13	38	61	188	–
ETHUSDC hfgrid	held out	−1.60	−1.02	+0.22	+1.86	+4.71	+0.31
ETHUSDC hfgrid	training	+1.93	+1.40	+3.75	+5.65	+11.47	+3.36
	fills/day held out	22	7	17	24	90	17
SUIUSDC OBI	held out	+2.52	+0.32	+0.83	+4.23	+8.99	+2.61
SUIUSDC OBI	training	+3.60	−2.65	+4.16	+10.94	+18.57	+3.21
	fills/day held out	175	438	544	599	715	9
XRPUSDT OBI	held out	+3.53	+3.69	+5.08	+5.25	+4.49	+3.64

configuration	sessions	as chosen	mid +0.5s	mid +1s	mid +1.5s	mid +3s	mid +1s, own signal off
XRPUSDT OBI	training	+4.79	+3.18	+5.06	+7.66	+16.10	+6.33
	fills/day held out	189	179	211	239	254	1
AVAXUSDC OBI	held out	+2.68	+9.73	+11.95	+15.16	+15.98	+1.14
AVAXUSDC OBI	training	+7.30	+9.66	+10.54	+10.00	+13.59	+4.32
	fills/day held out	150	106	112	116	138	2
BNBUSDC GLFT	held out	+1.72	+7.15	+14.72	+21.18	+41.33	–
BNBUSDC GLFT	training	+1.71	+6.11	+10.36	+14.98	+25.89	–
	fills/day held out	489	436	691	943	1704	–

Cumulative return, %. "own signal off": the oracle replaces the strategy's signal instead of adding to it.

- **The request's hypothesis holds**: centring on the mid 1 s ahead raises held-out return on seven of eight configurations (all but SUIUSDC), 3 s ahead on all eight, and the gain grows with the horizon. ETH GLFT turns from −3.25% to +4.30% at 1 s and +7.53% at 1.5 s; BNB GLFT gains 13 points and AVAX OBI 9 at 1 s.
- **It changes little on XRP** (+0.3 points at 1 s on XRPUSDC): XRP OBI quotes about 19 bp from the mid, and a second's move is usually under 1 bp, so knowing it barely moves the quotes relative to where fills happen.
- **SUIUSDC is worse with 0.5 s of foresight** (+0.32% against +2.52%) and better only from 1.5 s: moving the quotes by the coming small move trades more (438 fills a day against 175) without avoiding the moves that matter.
- **Switching OBI's own signal off costs almost every fill** (9–10 a day on ETH and SUI): OBI's quotes are too far out to fill without its signal pulling one side in. The signal matters for where fills happen, not for forecasting.

5. How good a forecast has to be

Between the mid ($R^2 = 0$) and the oracle ($R^2 = 1$): quotes centred on a **noisy oracle** — `r2 × (move + noise)`, the noise scaled to make the forecast's R^2 exactly `r2` and as autocorrelated as the move itself.

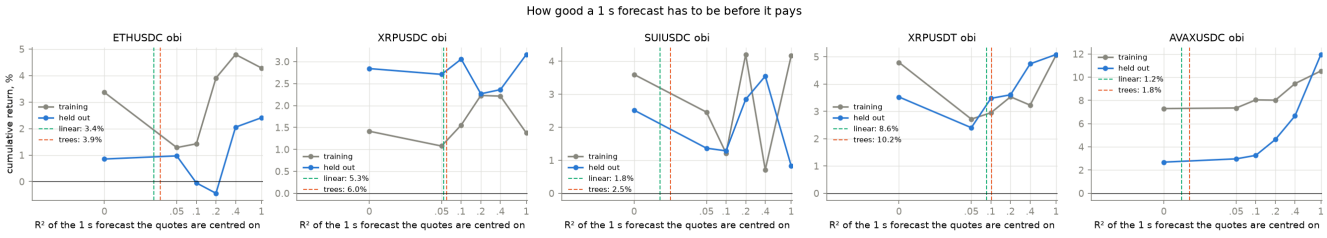


Figure — return against the R^2 of the 1 s forecast the quotes are centred on; dashed, the R^2 the real linear and boosted-tree forecasts reach on that asset.

configuration	best real forecast, 1 s	sessions	C8 choice	R^2 0.05	R^2 0.1	R^2 0.2	R^2 0.4	R^2 1 (the future mid)
ETHUSDC OBI	3.9%	held out	+0.85	+0.97	−0.05	−0.44	+2.05	+2.41
ETHUSDC OBI		training	+3.37	+1.28	+1.43	+3.91	+4.80	+4.29
XRPUSDC OBI	6.0%	held out	+2.84	+2.71	+3.06	+2.27	+2.36	+3.16
XRPUSDC OBI		training	+1.41	+1.08	+1.56	+2.23	+2.21	+1.37
SUIUSDC OBI	2.5%	held out	+2.52	+1.37	+1.29	+2.84	+3.55	+0.83
SUIUSDC OBI		training	+3.60	+2.46	+1.21	+4.20	+0.71	+4.16
XRPUSDT OBI	10.2%	held out	+3.53	+2.40	+3.48	+3.61	+4.74	+5.08
XRPUSDT OBI		training	+4.79	+2.72	+2.95	+3.53	+3.23	+5.06
AVAXUSDC OBI	1.8%	held out	+2.68	+2.97	+3.27	+4.64	+6.66	+11.95
AVAXUSDC OBI		training	+7.30	+7.35	+8.04	+8.02	+9.45	+10.54

Cumulative return, %; R^2 of a 1 s forecast.

- **Below $R^2 \approx 0.2$ a forecast mostly costs money.** ETH OBI loses 1.9–2.1 points on training sessions at R^2 0.05–0.1; SUIUSDC and XRPUSDT lose on both halves. XRPUSDC is the exception, level or slightly better at 0.1.
- **It takes $R^2 \approx 0.4$ to beat the C8 choice on both training and held-out sessions** on ETH OBI, and even that is not enough on SUIUSDC or XRPUSDT. AVAXUSDC is the exception: every R^2, even 0.05, is at least level.
- **The real forecasts sit at 0.02–0.10**, where the curve is flat or falling. Forecasting the mid is not the route to the oracle's gain.

6. Causal fair prices, chosen on training sessions

The curve is a mapping, not a verdict, so each real forecast was also tried as the fair price, like every other C9 variant: the micro-price, a fitted micro-price, the linear and the boosted-tree 1 s forecast, each added to the strategy's signal or replacing it, at ×1 and ×2; each family's best variant on training sessions reported held out and stress-tested.

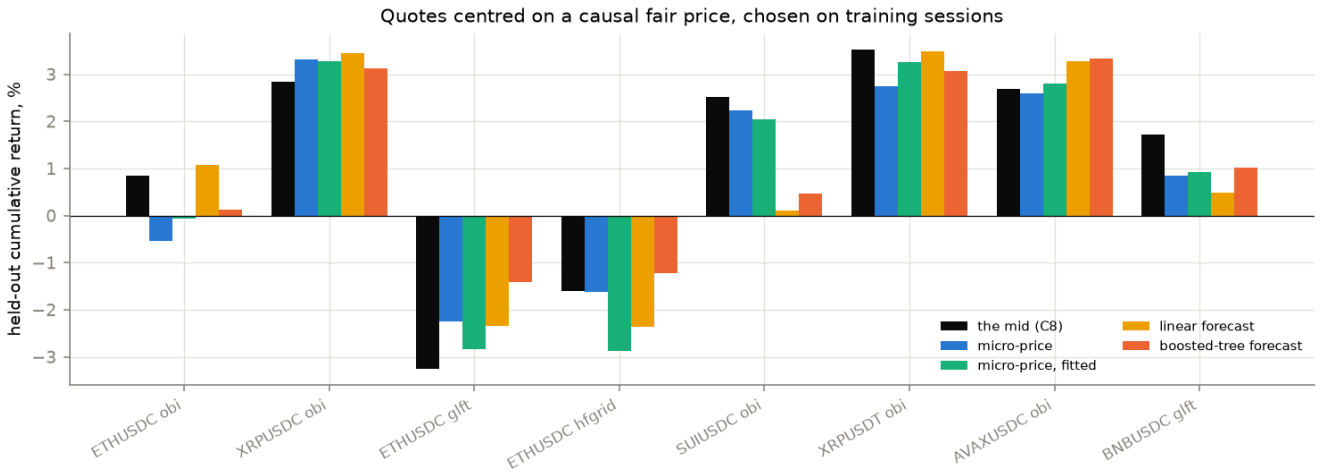


Figure — held-out return of each fair price's training-best variant.

configuration	fair price	chosen variant	training	held out	without 09-22	fills/day
ETHUSDC OBI	the mid (C8)	–	+3.37	+0.85	+0.14	369
	micro-price	feat:micro_bp, ×1, added to own signal	+3.37	−0.53	−0.58	411
	micro-price, fitted	pred:micro:10, ×2, added to own signal	+2.51	−0.06	−0.06	589
	linear forecast	pred:lin:10, ×1, added to own signal	−0.05	+1.08	+0.61	227
	boosted-tree forecast	pred:gbm:10, ×1, added to own signal	+1.79	+0.13	+0.02	248
XRPUSDC OBI	the mid (C8)	–	+1.41	+2.84	+0.09	106
	micro-price	feat:micro_bp, ×2, added to own signal	+1.69	+3.32	−0.10	109
	micro-price, fitted	pred:micro:10, ×2, added to own signal	+2.04	+3.28	+0.02	120
	linear forecast	pred:lin:10, ×2, added to own signal	+1.17	+3.45	+0.24	94
	boosted-tree forecast	pred:gbm:10, ×2, added to own signal	+0.14	+3.12	−0.54	105
ETHUSDC GLFT	the mid (C8)	–	+3.37	−3.25	−2.34	27
	micro-price	feat:micro_bp, ×1, added to own signal	+2.68	−2.25	−1.65	19
	micro-price, fitted	pred:micro:10, ×1, added to own signal	+2.12	−2.84	−2.25	20
	linear forecast	pred:lin:10, ×2, added to own signal	+4.89	−2.33	−2.23	17
	boosted-tree forecast	pred:gbm:10, ×2, added to own signal	+3.15	−1.41	−1.68	20

configuration	fair price	chosen variant	training	held out	without 09-22	fills/day
ETHUSDC hfgrid	the mid (C8)	–	+1.93	−1.60	−0.51	22
	micro-price	feat:micro_bp, ×2, added to own signal	+2.04	−1.61	−0.39	24
	micro-price, fitted	pred:micro:10, ×2, replaces own signal	+2.87	−2.87	−1.08	24
	linear forecast	pred:lin:10, ×2, replaces own signal	+2.99	−2.36	−0.97	27
	boosted-tree forecast	pred:gbm:10, ×2, replaces own signal	+2.62	−1.22	−0.46	22
SUIUSDC OBI	the mid (C8)	–	+3.60	+2.52	+1.77	175
	micro-price	feat:micro_bp, ×2, added to own signal	+5.22	+2.24	+1.19	239
	micro-price, fitted	pred:micro:10, ×1, added to own signal	+2.70	+2.05	+0.32	254
	linear forecast	pred:lin:10, ×2, added to own signal	+3.55	+0.10	−0.57	342
	boosted-tree forecast	pred:gbm:10, ×1, added to own signal	+3.26	+0.47	+0.08	186
XRPUSDT OBI	the mid (C8)	–	+4.79	+3.53	+0.48	189
	micro-price	feat:micro_bp, ×2, added to own signal	+5.42	+2.75	−0.32	189
	micro-price, fitted	pred:micro:10, ×1, added to own signal	+4.59	+3.27	+0.19	209
	linear forecast	pred:lin:10, ×1, added to own signal	+3.66	+3.48	+0.45	196
	boosted-tree forecast	pred:gbm:10, ×2, added to own signal	+4.67	+3.07	−0.45	178
AVAXUSDC OBI	the mid (C8)	–	+7.30	+2.68	+2.34	150
	micro-price	feat:micro_bp, ×2, added to own signal	+7.86	+2.59	+2.48	149
	micro-price, fitted	pred:micro:10, ×2, added to own signal	+7.26	+2.81	+2.59	149
	linear forecast	pred:lin:10, ×2, added to own signal	+7.65	+3.27	+3.15	149
	boosted-tree forecast	pred:gbm:10, ×2, added to own signal	+8.04	+3.33	+2.94	143
BNBUSDC GLFT	the mid (C8)	–	+1.71	+1.72	+1.53	489
	micro-price	feat:micro_bp, ×1, added to own signal	+2.09	+0.85	+0.27	502

configuration	fair price	chosen variant	training	held out	without 09-22	fills/day
	micro-price, fitted	pred:micro:10, ×2, added to own signal	+2.06	+0.93	+0.88	457
	linear forecast	pred:lin:10, ×2, added to own signal	+2.38	+0.49	+0.77	456
	boosted-tree forecast	pred:gbm:10, ×2, added to own signal	+2.20	+1.03	+1.23	517

Cumulative return, %. "chosen variant": the family's best on the 24 training sessions.

Counting only the variants that beat the C8 choice on training sessions — the ones a training-session selection would adopt:

- **AVAXUSDC OBI is the one real improvement**: the boosted-tree forecast gives +3.33% held out against +2.68%, +2.94% against +2.34% without the crash, and it survives removing the best session (+0.13% against −0.37%), a 1 bp maker fee (+2.35% against +1.69%) and 4× latency (+3.79%). Like the C8 choice it is fragile to a 20% narrower floor (+0.29%).
- **XRPUSDC OBI gains about 0.4 points with the micro-price** (+3.28% to +3.32% against +2.84%), and all of it is the crash: without 09-22 it is +0.02% against +0.09%.
- **The rest adopt a worse configuration**: SUIUSDC +2.24% against +2.52%, XRPUSDT +2.75% against +3.53%, BNB GLFT +0.49% to +1.03% against +1.72%, and the ETH GLFT and hfgrid variants still lose. On ETH OBI no fair price beats the C8 choice on training sessions (the micro-price ties it).

7. Acting only on large moves

The oracle's gain might come from a few large moves, and a forecast that is weak on average might still spot those. Two new actions: **widen** only the side the coming move would run over, by the move, or **pull** it — each on every step, or only when the move is in the top 10% or 1% of the training sessions' moves.

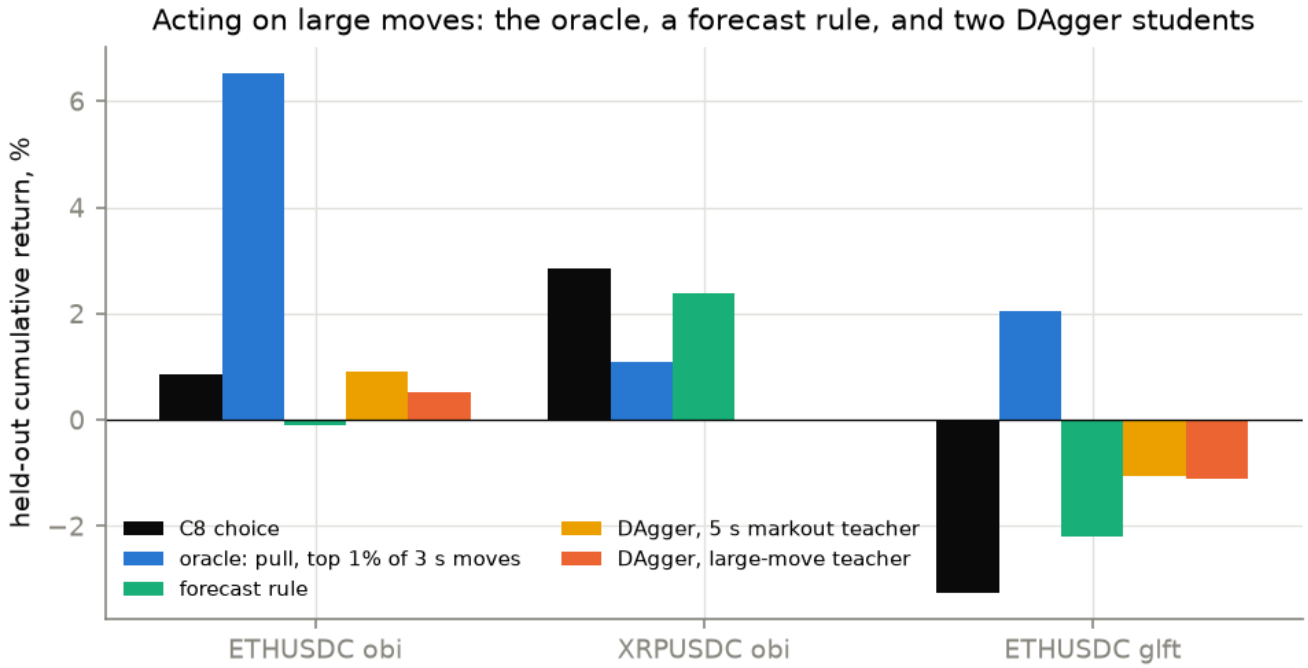


Figure — held-out return: the C8 choice, the best large-move oracle, the forecast rule chosen on training sessions, and the two DAgger students (§8).

configuration	sessions	C8 choice	oracle: pull, top 10% of 1 s moves	oracle: pull, top 1% of 3 s moves	oracle: widen, top 10% of 1 s moves	oracle: widen endangered side by the 1 s move	forecast rule (chosen on training)
ETHUSDC OBI	held out	+0.85	+2.98	+6.53	+2.06	+0.79	−0.10
ETHUSDC OBI	training	+3.37	+13.41	+11.68	+5.99	+5.71	+4.23
	fills/day held out	369	590	1171	600	700	287
XRPUSDC OBI	held out	+2.84	+1.89	+1.09	+3.28	+3.04	+2.38
XRPUSDC OBI	training	+1.41	+0.65	+0.37	+0.72	+1.71	+1.98
	fills/day held out	106	75	151	95	93	101
ETHUSDC GLFT	held out	−3.25	−0.70	+2.05	−0.54	−0.37	−2.20
ETHUSDC GLFT	training	+3.37	+2.24	+5.78	+0.74	+0.08	+4.10
	fills/day held out	27	3	41	4	4	15

Cumulative return, %. Oracle columns use the actual future mid; the forecast rule is the best on training sessions of the boosted-tree 1 s and 3 s forecasts gated at their top 5% or 1%, as a shift, a widening or a pull.

- **A perfect large-move detector would be enough on ETH**: pulling a quote before the top 1% of 3 s moves against it takes ETH OBI from +0.85% to +6.53% held out and ETH GLFT from −3.25% to +2.05%, acting on about 1% of steps. On XRP no oracle variant beats the C8 choice held out.
- **The forecasts cannot see those moves coming.** The top 1% of the boosted-tree forecasts expect 0.59–0.65 bp; the top 1% of actual moves are 2.9–3.3 bp at 1 s and 4.9–5.5 bp at 3 s. Gating on the forecast's extremes selects ordinary steps.
- **So the rules built on them do not help**: chosen on training sessions (where all three beat the C8 choice), they give −0.10% against +0.85% on ETH OBI, +2.38% against +2.84% on XRP OBI, and −2.20% against −3.25% on ETH GLFT — an improvement that still loses, on 15 fills a day.

8. DAgger with a privileged teacher

If the forecast of the move is the wrong target, learn the decision instead. **DAgger** trains a student policy on the states it visits itself, labelled by a teacher: here the teacher sees the future mid, and the student sees only what the quoter sees.

- **The student** decides, per side and step, whether to pull (or widen) the quote the rule is about to post. Its score is a boosted-tree model of market features only (the 17 of §3, mirrored so one model serves both sides), plus a logistic layer over the student's own state that only the kernel knows when it quotes: the quote's distance from the mid, the inventory a fill would add to, and that side's markout toxicity.
- **Rounds.** Round 0 labels the states the C8 rule visits on the 24 training sessions; each later round rolls out the current student, labels the states *it* visits, adds them and refits on everything so far. Only training sessions are rolled out or labelled. The student's threshold and action are then chosen on training sessions and stress-tested like any other variant.
- **Targets**: ETH GLFT and ETH OBI, the two configurations with the largest oracle gap.

Two teachers. The first asked, for each posted quote, whether it would be filled within 1 s (the opposite touch reaching its price) and whether that fill would lose on a 5 s markout after the rebate. It was misaligned: a quote the opposite touch reaches is mostly one the price runs through, so 58–69% of its labelled fills lose and the student learns to pull nearly everything (84% of GLFT quotes and 99–100% of OBI quotes at $p > 0.5$); and the strategies earn over longer holds than 5 s. Its result is reported as it came out. The second teacher is the policy §7 showed to pay: **pull a side just before a move against it in the top 1% of 3 s moves**. Its labels are rare (0.2–0.3% of steps), and its student learns to classify large moves directly rather than forecast their size.

configuration	student	rounds, labelled rows	teacher says pull	held-out AUC	chosen action	training	held out	without 09-22	fills/day
ETHUSDC GLFT	none (C8 choice)	–	–	–	–	+3.37	−3.25	−2.34	27

configuration	student	rounds, labelled rows	teacher says pull	held-out AUC	chosen action	training	held out	without 09-22	fills/day
	DAgger, 5 s markout teacher	3, 79,065	67.6%	–	pull (p > 0.7)	+5.02	−1.05	−0.60	19
	DAgger, large-move teacher	2, 4,510,122	0.5%	0.71	pull (p > 0.1)	+4.06	−1.12	−1.66	19
ETHUSDC OBI	none (C8 choice)	–	–	–	–	+3.37	+0.85	+0.14	369
	DAgger, 5 s markout teacher	3, 656,772	58.0%	–	widen 2 bp (p > 0.7)	+3.80	+0.91	+0.02	369
	DAgger, large-move teacher	2, 4,542,042	0.5%	0.71	widen 5 bp (p > 0.01)	+2.46	+0.52	+0.16	448

"teacher says pull": share of labelled weight (fills for the first teacher, steps for the second) on which the teacher pulls. "held-out AUC": on the last training session, which the tree model uses for early stopping only.

- **Both students improve ETH GLFT and neither makes it pay.** The markout teacher's student (pull at p > 0.7) gives −1.05% held out and the large-move teacher's (pull at p > 0.1) −1.12%, against −3.25% for the C8 choice, on 19 fills a day against 27. Both also beat the C8 choice on training sessions (+5.02% and +4.06% against +3.37%) and under most stresses; the large-move student is even positive at 4× latency (+2.37%) and with κ halved (+1.61%), but negative in every cost and session stress.
- **Neither helps ETH OBI.** The markout student's choice (widen 2 bp at p > 0.7) acts so rarely that it is the rule: +0.91% against +0.85%, +0.02% against +0.14% without the crash. The large-move student's best variant on training sessions (widen 5 bp at p > 0.01) earns less than the rule on training (+2.46% against +3.37%), so a training-session selection would not adopt it; held out it gives +0.52%.
- **Why the large-move student falls short of its teacher.** It ranks the teacher's pulls well above chance (AUC 0.71 on the held-back training session) but the events are rare: at p > 0.05 it acts on 0.3–0.7% of steps and is right 3–11% of the time. The teacher is worth +6.53% on ETH OBI because it is never wrong; a student that pulls ten times for every real move gives up more spread than it saves. Another DAgger round changes nothing here, since a student that rarely acts visits the same states as the rule.

How this differs from the RL tried before. C3, C4, C6 and C8 learned from reward — session or step PnL — and the effect of any one quoting decision is buried in the mark-to-market noise of the position, so the learners never found a signal. DAgger learns from a label at every step, which removes that problem entirely; what it cannot remove is the limit of what its inputs reveal about the teacher's privileged information. The students here reach that limit: they rank the dangerous moments (AUC 0.71) but cannot act on them precisely enough. The comparison across Part C, the clues that could make RL tractable, and whether the strategies' parameters are too rigid are in the Part C overview, §6.

9. Denoising on every asset

C8 tried three denoisers on ETHUSDC and XRPUSDC. Here five run on OBI on all nine assets and on BNB GLFT, each family's training-best variant reported held out and stress-tested — whether or not it looked better, so no denoiser is dismissed on the plain backtest alone:

family	variants	what it does
signal smoothing	1, 5, 20 s half-life	exponential average of the fair-price signal
alpha clipping	±1, ±1.5, ±2.5 σ	cap OBI's standardised imbalance
standardisation window	5, 30, 60 min (default 10)	the window OBI's imbalance is standardised over
imbalance depth	5, 20, 40 bp (default 10)	how deep into the book the imbalance sums
markout half-life	5, 10, 50 fills (default 20)	how many fills the toxicity average spans

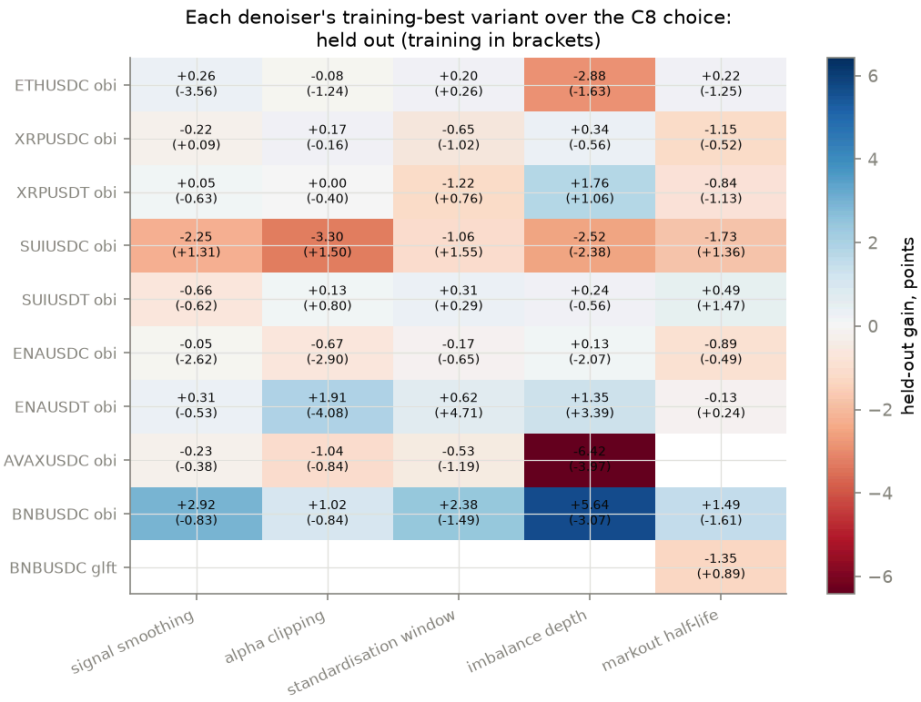


Figure — held-out gain of each denoiser's training-best variant over the C8 choice, points (training gain in brackets).

configuration	C8 choice, training / held out	fills/day	signal smoothing	alpha clipping	standardisation window	imbalance depth	markout half-life
ETHUSDC OBI	+3.37 / +0.85	369	−3.56 / +0.26	−1.24 / −0.08	+0.26 / +0.20	−1.63 / −2.88	−1.25 / +0.22
XRPUSDC OBI	+1.41 / +2.84	106	+0.09 / −0.22	−0.16 / +0.17	−1.02 / −0.65	−0.56 / +0.34	−0.52 / −1.15
XRPUSDT OBI	+4.79 / +3.53	189	−0.63 / +0.05	−0.40 / +0.00	+0.76 / −1.22	+1.06 / +1.76	−1.13 / −0.84
SUIUSDC OBI	+3.60 / +2.52	175	+1.31 / −2.25	+1.50 / −3.30	+1.55 / −1.06	−2.38 / −2.52	+1.36 / −1.73
SUIUSDT OBI	+3.65 / −0.53	44	−0.62 / −0.66	+0.80 / +0.13	+0.29 / +0.31	−0.56 / +0.24	+1.47 / +0.49
ENAUSDC OBI	+3.73 / +2.30	2	−2.62 / −0.05	−2.90 / −0.67	−0.65 / −0.17	−2.07 / +0.13	−0.49 / −0.89
ENAUSDT OBI	+12.36 / −1.56	6	−0.53 / +0.31	−4.08 / +1.91	+4.71 / +0.62	+3.39 / +1.35	+0.24 / −0.13
AVAXUSDC OBI	+7.30 / +2.68	150	−0.38 / −0.23	−0.84 / −1.04	−1.19 / −0.53	−3.97 / −6.42	–
BNBUSDC OBI	+3.04 / −3.40	524	−0.83 / +2.92	−0.84 / +1.02	−1.49 / +2.38	−3.07 / +5.64	−1.61 / +1.49
BNBUSDC GLFT	+1.71 / +1.72	489	–	–	–	–	+0.89 / −1.35

Gain over the C8 choice, points of return, training / held out. ENA OBI trades 2–6 times a day, so its rows are noise.

configuration	signal smoothing	alpha clipping	standardisation window	imbalance depth	markout half-life
ETHUSDC OBI	signal_hl 50	alpha_clip 2.5	obi_window 3000	obi_depth 0.002	markout_halflife 10
XRPUSDC OBI	signal_hl 50	alpha_clip 2.5	obi_window 36000	obi_depth 0.004	markout_halflife 10
XRPUSDT OBI	signal_hl 10	alpha_clip 2.5	obi_window 36000	obi_depth 0.004	markout_halflife 10
SUIUSDC OBI	signal_hl 200	alpha_clip 1.5	obi_window 3000	obi_depth 0.0005	markout_halflife 10
SUIUSDT OBI	signal_hl 10	alpha_clip 2.5	obi_window 3000	obi_depth 0.0005	markout_halflife 10
ENAUSDC OBI	signal_hl 10	alpha_clip 1.5	obi_window 18000	obi_depth 0.004	markout_halflife 5
ENAUSDT OBI	signal_hl 10	alpha_clip 2.5	obi_window 18000	obi_depth 0.002	markout_halflife 10

configuration	signal smoothing	alpha clipping	standardisation window	imbalance depth	markout half-life
AVAXUSDC OBI	signal_hl 10	alpha_clip 2.5	obi_window 3000	obi_depth 0.0005	–
BNBUSDC OBI	signal_hl 50	alpha_clip 2.5	obi_window 36000	obi_depth 0.004	markout_halflife 50
BNBUSDC GLFT	–	–	–	–	markout_halflife 5

- **The best configuration gets worse.** On SUIUSDC OBI every denoiser that helps the training sessions (by 1.3–1.6 points) costs 1–3.3 points held out: more parameters fit the training month.
- **The two gains that look real fail a stress test.** ETH OBI with a 5-minute window (+0.26 / +0.20) is worse than the C8 choice under a 1 bp fee (−2.09% against −1.42%); XRPUSDT OBI with a 40 bp imbalance band (+1.06 / +1.76, +5.29% held out) is robust to costs, but its gain is the crash: −0.83% without 09-22 against +0.48%.
- **SUIUSDT** improves on both halves with the markout window and clipping, and still loses held out. **BNB OBI** improves held out with every denoiser and on training with none, so no denoiser would be chosen for it.

The denoising that works is the one already inside the markout rule and the circuit breaker (C8 §4.3).

10. Stress tests

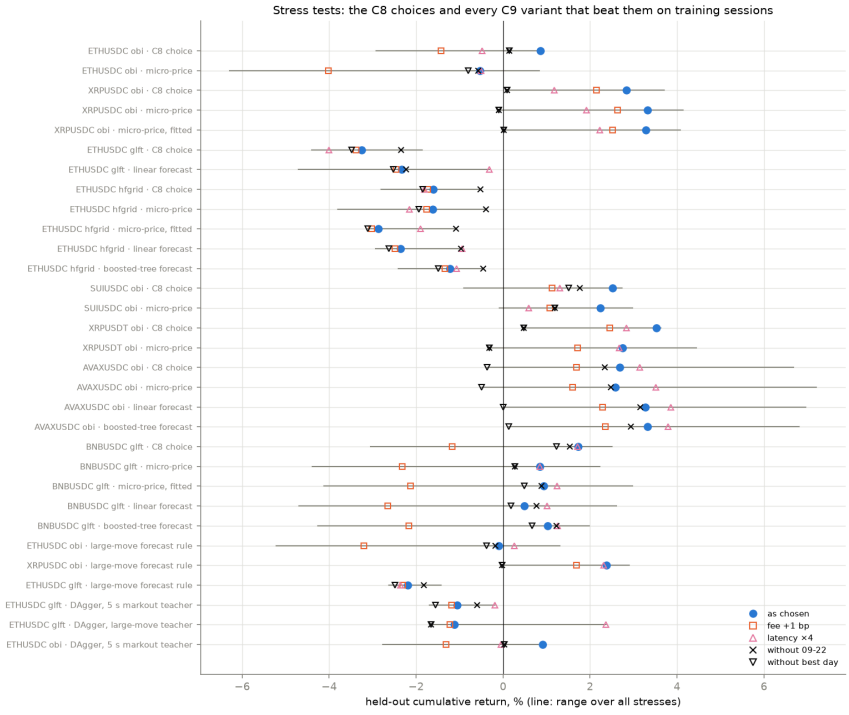


Figure — the C8 choices and every C9 variant that beat them on training sessions: held-out return as chosen, under a 1 bp maker fee, at 4× latency, and without the crash or the best session; the line spans all stresses.

The complete stress tables — every family's training-best variant on every configuration, including those that did not beat the C8 choice — are in the appendix. The pattern:

- **Nothing C9 adopts turns a cost-sensitive configuration robust.** ETH OBI and BNB GLFT lose under a 1 bp fee with every variant, as they did in C8.
- **AVAXUSDC OBI with the boosted-tree fair price** is the one adoption that is at least as good as the C8 choice under every stress, and better without its best session.
- **SUIUSDC OBI's C8 choice stays the most robust configuration in Part C**: no C9 variant beats it on training sessions and held out together.

11. So what?

- **The fair price the quotes are centred on matters enormously — if it is right.** One to three seconds of foresight is worth more, on most configurations, than any control or idea tried since C6.
- **Forecasting it is not the way there.** Real forecasts reach R^2 0.02–0.10; the gain starts near 0.4. Quoting around the micro-price or a forecast helps one asset (AVAXUSDC) and hurts or does nothing on the rest.
- **The value is concentrated in rare large moves**, which is why a perfect detector acting on 1% of steps captures most of it, and why a forecast of the average move cannot: it never predicts a large move.
- **Learning the decision instead of the forecast helps the strategy that loses and not the ones that make money.** Both DAgger students halve ETH GLFT's held-out loss (−3.25% to about −1.1%); neither beats the rule on ETH OBI. The large-move student can rank the dangerous moments (AUC 0.71), but not precisely enough to act on them profitably.
- **Denoising is done.** Across nine assets, no denoiser survives both the held-out week and the stress tests on a profitable configuration.

12. What would push further

- **The forward test** (still the one Part C follow-up not run): the frozen SUIUSDC and XRP OBI choices, and AVAXUSDC with the boosted-tree fair price, on sessions after 2025-09-25.
- **Information the book does not carry.** The large moves are what matters and the book does not anticipate them at 1–3 s. The candidates are outside it — the same asset's USDT perpetual, spot, other venues' perpetuals, liquidation feeds — and most are free: the full list, where each can be had, and how it would be tested without look-ahead are in the Part C overview, §7. C7 already showed BTCUSDT's book adds nothing at 1 s.
- **Tail-event classifiers with a longer label horizon**, trained on the whole cross-section of assets rather than one, since a top-1% move is rare on any single asset.

13. How it was run, and what went wrong

- **Where.** ETHUSDC and XRPUSDC on this machine (four cores, 16 GB), the other assets and some of the ETH/XRP stages on Kaggle CPU sessions (four cores, 30 GB), all writing the same (configuration, session) cache; the fitted models for ETH and XRP were embedded in the Kaggle notebooks so every host used the same ones.
- **Memory.** Three ETH workers again ran out of memory on the heavy 09-24 session, and a feature extraction beside three backtest workers did too; both were rerun with fewer workers.
- **Container restarts.** The machine restarted several times during the runs; every stage resumes from its cache, DAgger's rollouts from their files, so nothing was lost but time.
- **The first DAgger teacher** was misaligned (§8), found from its label statistics before its student was trusted, and replaced by the large-move teacher; both are reported.
- **Kaggle's 1 MB notebook limit** meant the embedded cache slice had to be cut to the configurations and fields each stage reads.

14. Sources

what	source
the month split, the stress set, the cache	C8
the strategies and the markout rule	C6, C7
earlier book-information study (per second)	C7 §5
micro-price	Stoikov, "The micro-price" (2018)
order-flow imbalance	Cont, Kukanov & Stoikov (2014)
DAgger	Ross, Gordon & Bagnell, "A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning" (2011)
code	`experiments/c9_fair_price/` (`c9.py`, `features.py`, `fair.py`, `predict.py`, `dagger.py`, `report.py`), `experiments/c6_strategy_zoo/zoo.py`

Appendix: every stress test

Held-out cumulative return, %, seven sessions, for each C8 choice and each family's training-best variant.

Fair prices

configuration	variant	held out	fills/day	ex 09-22	ex best day	fee 0	fee +1 bp	fee +2 bp	latency ×2	latency ×4	queue risk-averse	queue log	size ×2	floor ×0.8	floor ×1.25	κ ×0.5	κ ×2
ETHUSDC OBI	C8 choice	+0.85	369	+0.14	+0.14	+0.09	−1.42	−2.92	−0.40	−0.48	+0.91	+0.37	+0.64	−0.17	+0.26	−0.58	+0.34
ETHUSDC OBI	micro-price	−0.53	411	−0.58	−0.80	−1.70	−4.01	−6.29	−0.57	−0.51	−0.42	+0.84	−0.43	+0.10	+0.36	−0.77	+0.60
ETHUSDC OBI	micro-price, fitted	−0.06	589	−0.06	−0.40	−1.30	−3.74	−6.13	+0.05	−0.30	−0.05	−0.15	+1.17	+0.21	+1.58	−0.25	−0.58
ETHUSDC OBI	linear forecast	+1.08	227	+0.61	+0.61	+0.41	−0.92	−2.24	+1.00	−0.87	+1.39	+1.58	+0.06	+1.21	+0.14	−0.92	+1.36

configuration	variant	held out	fills/day	ex 09-22	ex best day	fee 0	fee +1 bp	fee +2 bp	latency ×2	latency ×4	queue risk-averse	queue log	size ×2	floor ×0.8	floor ×1.25	κ ×0.5	κ ×2
ETHUSDC OBI	boosted-tree forecast	+0.13	248	+0.02	−0.18	−0.56	−1.93	−3.29	+0.85	+0.79	+1.13	+0.83	−1.34	+0.22	+1.80	+0.78	−0.25
XRPUSDC OBI	C8 choice	+2.84	106	+0.09	+0.09	+2.61	+2.14	+1.68	+3.22	+1.17	+2.27	+2.49	+2.83	+3.71	+2.98	+3.57	+2.80
XRPUSDC OBI	micro-price	+3.32	109	−0.10	−0.10	+3.09	+2.63	+2.17	+2.95	+1.92	+3.77	+4.15	+3.41	+4.00	+2.88	+3.71	+2.28
XRPUSDC OBI	micro-price, fitted	+3.28	120	+0.02	+0.02	+3.02	+2.52	+2.01	+3.22	+2.23	+3.11	+3.05	+2.80	+3.30	+2.60	+4.08	+3.07
XRPUSDC OBI	linear forecast	+3.45	94	+0.24	+0.24	+3.24	+2.82	+2.41	+2.42	+1.34	+2.88	+2.57	+1.32	+2.70	+2.28	+3.15	+1.72
XRPUSDC OBI	boosted-tree forecast	+3.12	105	−0.54	−0.54	+2.89	+2.43	+1.97	+2.59	+1.90	+2.50	+2.59	+3.28	+3.92	+2.93	+3.60	+2.79
ETHUSDC GLFT	C8 choice	−3.25	27	−2.34	−3.49	−3.29	−3.38	−3.46	−3.78	−4.01	−3.25	−3.25	−3.88	−4.40	−1.97	−3.96	−1.86
ETHUSDC GLFT	micro-price	−2.25	19	−1.65	−2.52	−2.29	−2.38	−2.46	−3.74	−3.31	−2.25	−2.25	−4.49	−4.49	−3.31	−2.62	−0.88
ETHUSDC GLFT	micro-price, fitted	−2.84	20	−2.25	−3.19	−2.88	−2.96	−3.05	−3.13	−3.15	−2.84	−2.84	−3.64	−3.86	−2.23	−2.47	−2.97
ETHUSDC GLFT	linear forecast	−2.33	17	−2.23	−2.53	−2.37	−2.45	−2.54	−3.27	−0.31	−2.33	−2.33	−4.71	−3.67	−2.73	−2.60	−1.71
ETHUSDC GLFT	boosted-tree forecast	−1.41	20	−1.68	−1.68	−1.46	−1.56	−1.66	−1.33	−1.07	−1.41	−1.41	−1.72	−1.21	−1.43	−1.80	−1.27
ETHUSDC hfgrid	C8 choice	−1.60	22	−0.51	−1.85	−1.65	−1.73	−1.82	−1.74	−1.81	−1.60	−1.60	−2.80	−1.71	−2.00	−2.10	−1.83
ETHUSDC hfgrid	micro-price	−1.61	24	−0.39	−1.94	−1.66	−1.75	−1.85	−2.08	−2.16	−1.61	−1.68	−3.80	−1.85	−2.86	−2.15	−2.04
ETHUSDC hfgrid	micro-price, fitted	−2.87	24	−1.08	−3.11	−2.92	−3.02	−3.11	−2.22	−1.90	−2.87	−2.92	−2.85	−1.87	−2.43	−2.59	−2.08
ETHUSDC hfgrid	linear forecast	−2.36	27	−0.97	−2.62	−2.40	−2.49	−2.58	−1.24	−0.94	−2.36	−2.36	−2.56	−1.99	−2.93	−2.67	−1.71
ETHUSDC hfgrid	boosted-tree forecast	−1.22	22	−0.46	−1.48	−1.26	−1.34	−1.42	−1.73	−1.07	−1.22	−1.22	−2.34	−1.95	−2.05	−2.41	−1.79
SUIUSDC OBI	C8 choice	+2.52	175	+1.77	+1.52	+2.05	+1.13	+0.21	+1.86	+1.30	+2.46	+2.17	+2.74	+1.43	+1.36	−0.90	+0.56
SUIUSDC OBI	micro-price	+2.24	239	+1.19	+1.19	+1.85	+1.07	+0.30	+1.22	+0.59	+2.42	+2.66	+2.98	+1.02	+2.43	+1.35	−0.09
SUIUSDC OBI	micro-price, fitted	+2.05	254	+0.32	+0.32	+1.56	+0.59	−0.38	+3.30	+1.19	+1.68	+1.72	−2.33	+2.23	+2.75	+0.27	+0.56
SUIUSDC OBI	linear forecast	+0.10	342	−0.57	−0.57	−0.54	−1.80	−3.04	+2.68	+2.30	+3.22	+0.25	+0.12	+1.67	−0.58	−0.77	+2.50
SUIUSDC OBI	boosted-tree forecast	+0.47	186	+0.08	−0.04	−0.05	−1.09	−2.11	+1.15	+4.74	+0.40	+0.55	+1.33	+2.12	+2.82	+1.58	+0.38
XRPUSDT OBI	C8 choice	+3.53	189	+0.48	+0.48	+3.17	+2.45	+1.74	+3.09	+2.84	+2.67	+2.52	+3.63	+3.21	+1.70	+2.63	+2.76
XRPUSDT OBI	micro-price	+2.75	189	−0.32	−0.32	+2.40	+1.71	+1.02	+3.34	+2.67	+2.83	+3.25	+4.45	+2.35	+3.03	+3.27	+3.14
XRPUSDT OBI	micro-price, fitted	+3.27	209	+0.19	+0.19	+2.91	+2.21	+1.51	+3.40	+2.84	+2.69	+3.52	+3.47	+2.89	+3.27	+0.57	+2.54
XRPUSDT OBI	linear forecast	+3.48	196	+0.45	+0.45	+3.14	+2.46	+1.78	+3.06	+4.32	+1.64	+1.98	+4.18	+3.71	+2.57	+2.51	+1.98
XRPUSDT OBI	boosted-tree forecast	+3.07	178	−0.45	−0.45	+2.78	+2.19	+1.61	+3.13	+4.13	+2.60	+2.51	+4.34	+2.41	+2.61	+1.34	+2.62
AVAXUSDC OBI	C8 choice	+2.68	150	+2.34	−0.37	+2.35	+1.69	+1.04	+2.79	+3.14	+2.63	+2.67	+3.38	+0.25	+6.68	–	–
AVAXUSDC OBI	micro-price	+2.59	149	+2.48	−0.50	+2.26	+1.60	+0.95	+2.84	+3.51	+3.12	+3.33	+3.15	+0.96	+7.21	–	–
AVAXUSDC OBI	micro-price, fitted	+2.81	149	+2.59	−0.27	+2.48	+1.81	+1.15	+3.05	+2.98	+2.63	+2.97	+3.19	+0.36	+7.18	–	–
AVAXUSDC OBI	linear forecast	+3.27	149	+3.15	+0.01	+2.94	+2.29	+1.64	+3.01	+3.86	+3.16	+3.39	+3.59	+0.41	+6.96	–	–
AVAXUSDC OBI	boosted-tree forecast	+3.33	143	+2.94	+0.13	+3.00	+2.35	+1.71	+3.20	+3.79	+3.33	+3.52	+4.80	+0.29	+6.81	–	–
BNBUSDC GLFT	C8 choice	+1.72	489	+1.53	+1.23	+0.75	−1.17	−3.05	+0.58	+1.71	+1.53	+0.64	+2.50	+0.96	+1.05	+1.56	+1.01
BNBUSDC GLFT	micro-price	+0.85	502	+0.27	+0.27	−0.22	−2.32	−4.39	+0.86	+0.85	+0.91	+0.61	+2.22	+0.76	+0.61	+1.05	+0.97
BNBUSDC GLFT	micro-price, fitted	+0.93	457	+0.88	+0.49	−0.10	−2.12	−4.12	+1.18	+1.25	+1.18	+1.17	+2.98	+0.36	+1.05	+0.71	+0.60

configuration	variant	held out	fills/day	ex 09-22	ex best day	fee 0	fee +1 bp	fee +2 bp	latency ×2	latency ×4	queue risk-averse	queue log	size ×2	floor ×0.8	floor ×1.25	κ ×0.5	κ ×2
BNBUSDC GLFT	linear forecast	+0.49	456	+0.77	+0.18	−0.56	−2.65	−4.69	+1.15	+1.01	+0.47	+1.01	+2.61	+0.35	+0.64	+1.38	+0.16
BNBUSDC GLFT	boosted-tree forecast	+1.03	517	+1.23	+0.67	−0.05	−2.17	−4.26	+1.41	+1.26	+0.91	+0.91	+1.98	+0.97	+1.16	+1.15	+1.52

Denoising

configuration	variant	held out	fills/day	ex 09-22	ex best day	fee 0	fee +1 bp	fee +2 bp	latency ×2	latency ×4	queue risk-averse	queue log	size ×2	floor ×0.8	floor ×1.25	κ ×0.5	κ ×2
ETHUSDC OBI	C8 choice	+0.85	369	+0.14	+0.14	+0.09	−1.42	−2.92	−0.40	−0.48	+0.91	+0.37	+0.64	−0.17	+0.26	−0.58	+0.34
ETHUSDC OBI	markout half-life	+1.07	306	+0.89	+0.80	+0.36	−1.05	−2.44	+0.64	−0.51	+1.22	+1.08	+0.16	+1.42	+1.34	−0.91	+0.59
ETHUSDC OBI	signal smoothing	+1.11	220	+1.26	+0.59	+0.62	−0.34	−1.30	−0.13	+0.37	+1.49	+1.07	+0.24	+0.54	+1.07	+0.04	+1.44
ETHUSDC OBI	alpha clipping	+0.77	259	+0.14	+0.14	+0.09	−1.25	−2.58	+0.76	+0.59	+0.83	+0.46	+0.78	+0.01	+0.30	−0.58	+0.21
ETHUSDC OBI	standardisation window	+1.06	272	+0.32	+0.32	−0.00	−2.09	−4.16	+0.27	−0.51	+0.77	+0.69	+0.50	+0.32	+0.63	−0.72	+0.15
ETHUSDC OBI	imbalance depth	−2.02	205	−0.17	−2.36	−2.75	−4.19	−5.61	−1.16	−2.02	−2.71	−2.64	−3.28	+0.49	+0.65	−2.03	+1.32
XRPUSDC OBI	C8 choice	+2.84	106	+0.09	+0.09	+2.61	+2.14	+1.68	+3.22	+1.17	+2.27	+2.49	+2.83	+3.71	+2.98	+3.57	+2.80
XRPUSDC OBI	markout half-life	+1.69	112	−0.29	−0.29	+1.45	+0.97	+0.48	+2.93	+1.90	+2.12	+2.09	+2.97	+2.68	+1.60	+3.31	+2.02
XRPUSDC OBI	signal smoothing	+2.62	59	−0.05	−0.05	+2.49	+2.22	+1.95	+3.25	+3.16	+2.65	+2.61	+2.85	+3.49	+2.88	+3.42	+2.83
XRPUSDC OBI	alpha clipping	+3.01	102	−0.25	−0.25	+2.74	+2.20	+1.65	+3.34	+1.50	+2.79	+2.74	+3.79	+2.62	+2.63	+3.76	+3.03
XRPUSDC OBI	standardisation window	+2.19	67	−0.24	−0.24	+2.01	+1.63	+1.25	+3.43	+2.15	+2.29	+2.54	+2.88	+2.28	+3.23	+3.25	+1.83
XRPUSDC OBI	imbalance depth	+3.19	124	+0.28	+0.28	+2.93	+2.43	+1.92	+2.79	+2.95	+2.70	+3.09	+3.78	+3.51	+2.49	+3.74	+0.13
XRPUSDT OBI	C8 choice	+3.53	189	+0.48	+0.48	+3.17	+2.45	+1.74	+3.09	+2.84	+2.67	+2.52	+3.63	+3.21	+1.70	+2.63	+2.76
XRPUSDT OBI	markout half-life	+2.68	179	−0.42	−0.42	+2.38	+1.77	+1.16	+3.05	+1.62	+3.00	+2.53	+3.53	+2.80	+3.05	+3.60	+2.96
XRPUSDT OBI	signal smoothing	+3.58	215	+0.55	+0.55	+3.11	+2.19	+1.26	+3.09	+2.11	+1.86	+3.27	+3.54	+2.66	+2.39	+1.47	+3.51
XRPUSDT OBI	alpha clipping	+3.53	189	+0.48	+0.48	+3.17	+2.45	+1.74	+3.09	+2.92	+2.67	+2.52	+3.80	+3.21	+1.70	+2.63	+2.76
XRPUSDT OBI	standardisation window	+2.31	190	−0.31	−0.31	+1.99	+1.35	+0.71	+2.74	+2.76	+1.58	+2.29	+3.58	+1.94	+2.31	+0.94	+2.16
XRPUSDT OBI	imbalance depth	+5.29	334	−0.83	−0.83	+4.57	+3.14	+1.73	+5.35	+5.03	+6.14	+6.44	+6.55	+4.90	+5.57	+5.53	+3.97
SUIUSDC OBI	C8 choice	+2.52	175	+1.77	+1.52	+2.05	+1.13	+0.21	+1.86	+1.30	+2.46	+2.17	+2.74	+1.43	+1.36	−0.90	+0.56
SUIUSDC OBI	markout half-life	+0.79	200	+0.49	+0.28	+0.40	−0.38	−1.15	+0.34	+4.49	+1.22	+1.02	+0.98	+5.13	+2.06	+2.69	−2.39
SUIUSDC OBI	signal smoothing	+0.26	69	−0.40	−0.40	+0.13	−0.13	−0.38	+1.02	+1.39	+0.49	+0.48	−0.76	+0.18	+0.65	+0.74	+0.47
SUIUSDC OBI	alpha clipping	−0.78	131	−0.70	−1.20	−1.18	−1.98	−2.77	+1.33	+1.58	−0.32	−0.42	+1.04	+0.64	+1.57	−0.01	−2.28
SUIUSDC OBI	standardisation window	+1.46	242	−0.24	−0.24	+0.95	−0.07	−1.08	+3.06	+2.43	−0.02	+0.73	+0.29	−0.44	+3.45	−1.76	+1.05
SUIUSDC OBI	imbalance depth	−0.00	557	−0.64	−0.64	−0.85	−2.51	−4.16	+0.25	+1.58	−0.19	+0.00	−0.45	+0.09	+0.17	−4.76	+1.24
SUIUSDT OBI	C8 choice	−0.53	44	+1.90	−1.35	−0.61	−0.78	−0.95	−0.65	+1.76	−0.59	−0.35	+1.63	−0.64	−1.64	+0.21	−1.10
SUIUSDT OBI	markout half-life	−0.04	43	+2.57	−1.29	−0.14	−0.34	−0.55	−0.77	+0.01	−0.87	−0.82	−0.48	−0.88	−2.21	+0.67	+1.29
SUIUSDT OBI	signal smoothing	−1.18	42	+2.05	−2.21	−1.26	−1.42	−1.58	+1.27	+2.17	−1.75	−1.59	−1.77	+0.36	+0.01	−0.48	−0.51
SUIUSDT OBI	alpha clipping	−0.39	34	+1.01	−1.26	−0.45	−0.56	−0.68	−0.33	+1.90	+1.25	−0.38	−0.53	−0.21	−0.33	−0.16	−0.40
SUIUSDT OBI	standardisation window	−0.22	43	+1.53	−0.74	−0.29	−0.43	−0.57	−0.37	−0.16	−0.40	−0.71	−0.43	+1.07	+0.04	+2.41	+0.10
SUIUSDT OBI	imbalance depth	−0.29	26	+1.47	−0.75	−0.35	−0.48	−0.60	+0.51	+1.83	−0.65	−0.45	−0.59	+0.84	−1.15	−0.26	−0.30
ENAUSDC OBI	C8 choice	+2.30	2	+0.68	+0.68	+2.28	+2.26	+2.24	+1.34	+1.28	+2.05	+2.57	+2.70	+0.27	+0.42	+1.64	+1.74
ENAUSDC OBI	markout half-life	+1.41	2	+0.68	+0.68	+1.40	+1.38	+1.36	+1.34	+1.09	+1.17	+1.20	+1.24	−0.61	+0.01	+1.70	+1.02
ENAUSDC OBI	signal smoothing	+2.24	2	+0.84	+0.84	+2.23	+2.21	+2.20	+1.96	+1.74	+2.00	+2.40	+3.31	+0.18	+1.09	+2.03	+1.38
ENAUSDC OBI	alpha clipping	+1.63	2	+0.28	+0.28	+1.62	+1.60	+1.58	+1.30	+1.74	+1.62	+1.66	+1.26	+1.20	+1.62	+2.00	+1.56
ENAUSDC OBI	standardisation window	+2.13	3	+0.55	+0.55	+2.12	+2.09	+2.07	+1.37	+1.52	+1.89	+1.81	+2.00	−0.12	−0.15	+1.62	+1.64
ENAUSDC OBI	imbalance depth	+2.42	4	+1.18	+1.18	+2.41	+2.39	+2.37	+2.10	+3.19	+2.35	+2.70	+2.84	+3.16	+1.38	+2.46	+2.42
ENAUSDT OBI	C8 choice	−1.56	6	−1.13	−1.70	−1.58	−1.61	−1.65	−1.41	−1.14	−1.70	−2.09	−6.48	−0.76	−0.07	−2.93	−1.87
ENAUSDT OBI	markout half-life	−1.70	8	−0.59	−1.96	−1.71	−1.75	−1.78	−1.26	−1.13	−2.35	−2.63	−3.44	+0.76	+0.35	−1.94	−1.25
ENAUSDT OBI	signal smoothing	−1.25	7	−0.97	−1.40	−1.27	−1.31	−1.34	−1.14	−1.18	−0.67	−1.33	−6.43	+0.13	+1.11	−2.46	−1.17
ENAUSDT OBI	alpha clipping	+0.35	3	+0.03	+0.03	+0.34	+0.31	+0.29	+0.54	+0.43	+0.46	+0.33	−0.83	+0.91	+1.15	+0.14	+0.62
ENAUSDT OBI	standardisation window	−0.94	9	−0.69	−1.29	−0.96	−0.99	−1.03	−0.65	−0.58	−0.94	−0.69	−0.91	+2.53	−0.65	−4.50	−1.92

configuration	variant	held out	fills/day	ex 09-22	ex best day	fee 0	fee +1 bp	fee +2 bp	latency ×2	latency ×4	queue risk-averse	queue log	size ×2	floor ×0.8	floor ×1.25	κ ×0.5	κ ×2
ENAUSDT OBI	imbalance depth	−0.21	7	+0.98	−1.82	−0.23	−0.27	−0.30	−1.37	+0.56	+0.21	−1.57	+2.95	+1.18	+0.70	−1.17	−0.85
AVAXUSDC OBI	C8 choice	+2.68	150	+2.34	−0.37	+2.35	+1.69	+1.04	+2.79	+3.14	+2.63	+2.67	+3.38	+0.25	+6.68	–	–
AVAXUSDC OBI	signal smoothing	+2.46	125	+3.66	−0.55	+2.23	+1.78	+1.32	+2.46	+2.95	+2.72	+2.61	+2.61	+0.59	+7.55	–	–
AVAXUSDC OBI	alpha clipping	+1.64	145	+2.10	−0.81	+1.38	+0.84	+0.31	+1.59	+2.43	+1.30	+1.41	+1.07	+0.26	+1.02	–	–
AVAXUSDC OBI	standardisation window	+2.16	155	+2.14	−0.59	+1.82	+1.14	+0.46	+2.12	+2.67	+2.33	+2.34	+2.06	−0.07	+7.52	–	–
AVAXUSDC OBI	imbalance depth	−3.74	267	−2.09	−4.06	−4.19	−5.07	−5.96	−2.81	−1.99	−3.89	−3.88	−3.18	−1.03	−0.96	–	–
BNBUSDC OBI	C8 choice	−3.40	524	−2.67	−3.67	−4.47	−6.57	−8.64	−3.12	−1.45	−2.03	−3.82	−2.32	−1.23	−0.05	−1.40	−3.33
BNBUSDC OBI	markout half-life	−1.91	542	−2.16	−2.23	−3.09	−5.41	−7.69	−0.58	−1.69	−1.59	−1.94	−1.35	−2.17	−1.46	+0.26	−2.54
BNBUSDC OBI	signal smoothing	−0.48	340	−0.47	−0.90	−1.05	−2.19	−3.32	+0.46	−1.03	−0.87	−0.40	−1.59	−0.25	+0.20	−0.55	−1.74
BNBUSDC OBI	alpha clipping	−2.38	524	−2.29	−2.66	−3.43	−5.48	−7.50	−3.14	−1.59	−1.73	−2.93	−2.07	−1.50	−0.12	−2.28	−1.52
BNBUSDC OBI	standardisation window	−1.02	446	−1.13	−1.57	−1.81	−3.36	−4.89	−1.33	−0.49	−0.58	−0.18	−1.43	+0.58	+0.49	−1.95	−0.93
BNBUSDC OBI	imbalance depth	+2.23	668	+3.00	+1.09	+1.08	−1.21	−3.45	+1.29	+1.69	+0.66	+0.73	+0.03	+2.11	+1.53	−0.88	+0.73
BNBUSDC GLFT	C8 choice	+1.72	489	+1.53	+1.23	+0.75	−1.17	−3.05	+0.58	+1.71	+1.53	+0.64	+2.50	+0.96	+1.05	+1.56	+1.01
BNBUSDC GLFT	markout half-life	+0.37	358	+0.08	−0.15	−0.44	−2.03	−3.61	−0.44	−0.21	−1.25	−0.63	+1.70	−0.06	+0.05	+1.73	+0.97

Large moves and DAgger

configuration	variant	held out	fills/day	ex 09-22	ex best day	fee 0	fee +1 bp	fee +2 bp	latency ×2	latency ×4	queue risk-averse	queue log	size ×2	floor ×0.8	floor ×1.25	κ ×0.5	κ ×2
ETHUSDC OBI	C8 choice	+0.85	369	+0.14	+0.14	+0.09	−1.42	−2.92	−0.40	−0.48	+0.91	+0.37	+0.64	−0.17	+0.26	−0.58	+0.34
ETHUSDC OBI	large-move forecast rule	−0.10	287	−0.18	−0.37	−1.14	−3.20	−5.22	+0.17	+0.26	+0.14	−0.06	+1.17	+0.70	+1.30	−1.03	−0.06
XRPUSDC OBI	C8 choice	+2.84	106	+0.09	+0.09	+2.61	+2.14	+1.68	+3.22	+1.17	+2.27	+2.49	+2.83	+3.71	+2.98	+3.57	+2.80
XRPUSDC OBI	large-move forecast rule	+2.38	101	−0.02	−0.02	+2.15	+1.69	+1.24	+1.69	+2.32	+1.68	+1.62	+1.80	+2.91	+2.11	+2.71	+2.05
ETHUSDC GLFT	C8 choice	−3.25	27	−2.34	−3.49	−3.29	−3.38	−3.46	−3.78	−4.01	−3.25	−3.25	−3.88	−4.40	−1.97	−3.96	−1.86
ETHUSDC GLFT	large-move forecast rule	−2.20	15	−1.82	−2.49	−2.23	−2.31	−2.38	−1.57	−2.36	−2.20	−2.20	−2.63	−2.45	−2.53	−1.42	−1.91
ETHUSDC GLFT	DAgger, 5 s markout teacher	−1.05	19	−0.60	−1.56	−1.09	−1.18	−1.26	−0.77	−0.19	−1.05	−1.05	−0.64	−1.25	−0.50	−0.65	−1.69
ETHUSDC GLFT	DAgger, large-move teacher	−1.12	19	−1.66	−1.66	−1.15	−1.22	−1.29	−0.09	+2.37	−1.12	−1.12	+0.14	−1.52	−0.98	+1.61	−1.25
ETHUSDC OBI	DAgger, 5 s markout teacher	+0.91	369	+0.02	+0.02	+0.17	−1.31	−2.76	+0.45	−0.05	+0.52	+0.57	−0.34	−0.01	+0.81	−0.59	−0.20
ETHUSDC OBI	DAgger, large-move teacher	+0.52	448	+0.16	+0.16	−0.27	−1.83	−3.38	+0.15	+0.28	+0.03	+0.16	−0.26	+0.27	−0.40	+0.12	+0.52